

MASLYANINO, Russian Federation

WMO: 297360

Lat: 54.336N

Lon: 84.208E

Elev: 653

StdP: 14.35

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.2	-31.7	-43.2	0.5	-37.8	-37.0	0.7	-31.5	23.4	23.1	21.1	21.3	0.3	0	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	84.8	66.0	81.8	64.9	78.8	63.5	69.1	79.0	67.3	77.4	65.7	75.7	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	97.2	73.7	63.8	90.7	71.7	62.0	85.0	69.9	33.8	79.2	32.2	77.2	31.0	75.9	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.9	16.7	14.3	DB	-42.7	89.5	7.0	2.8	-47.8	91.5	-51.8	93.1	-55.7	94.7	-60.8	96.7	
			WB	-42.7	72.7	6.8	2.5	-47.6	74.6	-51.6	76.0	-55.4	77.5	-60.4	79.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.0	-1.9	4.4	19.9	37.0	50.9	61.4	65.1	60.1	48.8	36.6	19.1	4.7
	DBStd	25.56	16.33	14.01	11.59	9.49	8.82	7.10	5.38	6.26	7.35	8.96	14.16	16.11
	HDD50	7197	1609	1278	933	402	100	8	0	3	110	422	926	1406
	HDD65	11455	2074	1698	1398	841	441	145	66	176	487	882	1376	1871
	CDD50	1356	0	0	0	11	128	351	469	317	74	6	0	0
	CDD65	137	0	0	0	0	4	37	70	25	1	0	0	0
	CDH74	2007	0	0	0	6	187	608	752	400	54	0	0	0
	CDH80	462	0	0	0	0	40	137	175	97	13	0	0	0
Wind	WSAvg	5.1	4.6	5.1	5.4	5.7	5.9	4.8	3.9	3.9	4.3	5.5	5.9	5.6
Precipitation	PrecAvg	18.9	0.9	0.8	0.8	1.1	1.7	2.4	2.7	2.2	1.7	1.7	1.6	1.3
	PrecMax	23.6	1.9	1.9	1.9	2.6	3.8	4.8	5.8	3.8	3.5	4.0	3.9	2.8
	PrecMin	13.7	0.2	0.0	0.1	0.2	0.4	0.7	0.9	0.4	0.6	0.8	0.6	0.4
	PrecStd	3.0	0.4	0.4	0.4	0.5	1.0	1.1	1.3	0.9	0.7	0.7	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	34.6	48.2	72.8	85.0	87.3	88.6	88.0	80.7	66.5	48.7	35.4
		MCWB	32.3	32.4	38.9	54.4	61.2	65.6	67.6	67.3	60.6	52.4	44.0	32.8
	2%	DB	30.1	31.3	42.9	65.6	78.9	83.7	84.5	82.6	73.1	60.5	43.8	32.4
		MCWB	28.7	28.8	36.3	50.6	58.4	64.5	67.8	65.9	58.2	48.7	39.3	30.6
	5%	DB	26.0	28.3	39.1	60.0	74.8	80.7	81.4	78.4	68.3	55.4	39.1	29.6
		MCWB	24.4	26.2	34.1	47.7	56.3	64.1	66.5	63.7	55.7	46.6	36.1	27.8
	10%	DB	21.8	24.9	35.8	53.5	69.6	77.3	78.2	74.7	63.4	50.5	35.6	26.3
		MCWB	20.3	23.1	32.2	44.1	54.5	62.9	65.3	62.1	53.4	44.4	33.4	24.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	33.2	39.9	55.6	63.3	69.8	72.1	70.1	62.7	53.3	44.5	33.3
		MCDB	34.2	34.2	46.6	69.3	79.3	80.6	81.9	80.9	73.8	62.6	48.0	34.8
	2%	WB	28.7	29.2	36.8	52.1	60.8	67.2	70.1	67.5	59.5	50.0	39.6	30.8
		MCDB	30.1	30.8	41.5	63.9	75.8	78.7	79.6	78.5	70.8	57.9	42.7	32.3
	5%	WB	24.4	26.3	34.7	48.3	58.0	65.6	68.3	65.5	57.2	47.4	36.4	27.9
		MCDB	25.8	28.1	38.3	57.5	70.3	77.4	77.6	76.2	66.7	54.3	39.0	29.4
	10%	WB	20.4	23.2	32.4	44.8	55.6	63.9	66.6	63.4	54.5	44.7	33.4	24.8
		MCDB	21.9	24.9	35.8	53.4	67.8	74.4	75.5	72.6	61.4	50.4	35.3	26.4
Mean Daily Temperature Range	5% DB	MDBR	18.8	21.9	22.0	20.8	24.3	22.8	20.8	22.4	21.6	17.1	14.0	16.4
		MCDBR	15.0	18.9	23.1	32.7	34.6	30.5	27.0	29.7	32.5	28.1	13.8	14.7
		MCWBR	14.7	17.5	16.6	18.7	16.1	13.2	11.8	13.8	17.6	17.0	10.5	14.1
	5% WB	MCDBR	15.4	18.3	19.5	30.1	30.1	25.6	22.6	25.8	28.9	25.3	13.3	14.4
MCWBR		14.9	17.4	14.8	18.2	15.5	12.0	11.3	12.8	16.3	16.4	10.7	13.9	
Clear-Sky Solar Irradiance	taub	0.298	0.334	0.384	0.414	0.413	0.413	0.438	0.414	0.377	0.360	0.336	0.304	
	taud	1.995	1.939	1.873	2.035	2.177	2.263	2.194	2.246	2.293	2.207	2.014	1.940	
	Ebn at Noon	189	224	240	255	264	265	255	253	248	218	170	153	
	Edh at Noon	24	36	48	47	43	40	43	38	32	27	23	21	
All-Sky Solar Radiation	RadAvg	316	646	1097	1467	1702	1931	1742	1465	1005	508	256	206	
	RadStd	28	40	96	73	189	148	112	104	112	68	44	32	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page